

IIT Pokerbots 2026 : Finals

Problem Statement

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Anaconda Hold'em

1 Overview

The poker variant for the final round of IIT Pokerbots 2026 is “Anaconda Hold'em”. It is considerably different from the standard Texas Hold'em! In this game, each of the two players are dealt **7 hole cards**, while there are **no community cards** at all. At specific stages of the game - as described later - the players **exchange** a certain number of cards with each other, in an effort to form a better hand than the opponent. At showdown, each player forms the best possible five-card hand using any combination of their seven hole cards to determine the winner of the pot.

2 Glossary

- **Hole Cards:** The two private cards dealt to each player.
- **Pot:** The total chips accumulated from bets, blinds, and bids in a hand, to be claimed by the winner of the round.
- **Chips:** A player's individual resources they use for bets and other payments during a round.
- **Wager:** A player's contribution to the pot in the current betting round.
- **Small Blind:** The smaller forced bet posted at the start of a hand to initiate the pot.
- **Big Blind:** The larger forced bet posted at the start of a hand.
- **Call:** Match the current bet to remain in the hand.
- **Cost to Call:** Chips required to match the current bet to remain in the hand.
- **Check:** A “pass” or bet of 0.
- **Fold:** Withdraw from the hand and let other player claim the pot.
- **Bid:** The amount submitted in the Sneak Peek auction; the higher bidder pays the lower bid to the pot.
- **Showdown:** The stage when players reveal their hole cards to determine the winner of the round, who claims the pot.

3 Game Format

A game of **Anaconda Hold'em** consists of multiple rounds played between two players. In each round, both players are allocated a fixed number of **chips** before any cards are dealt. At the end of each round, the change in a player's chips is added to their cumulative **bankroll**, which **starts at zero**. After all rounds are completed, the player with the **highest cumulative bankroll** is **declared the winner of the game**.

The parameters used in the competition setting are as follows:

- **Number of rounds:** 1000
- **Chips allocated per round:** 5000
- **Small blind:** 10
- **Big blind:** 20

4 Sequence of Play

Each round of Anaconda Hold'em proceeds in the following order:

1. The **Small Blind** pays 10 chips and the **Big Blind** pays 20 chips to the pot.
2. Each player is dealt **seven** private hole cards.
3. Each player gives **three** of their cards to the opponent.
4. One round of betting takes place.
5. Each player gives **two** of their cards to the opponent.
6. One round of betting takes place.
7. Each player gives **one** of their cards to the opponent.
8. One final round of betting takes place.
9. If both players remain in the hand, a **showdown** occurs, and the player with the best five-card poker hand wins the pot.

5 Blinds

At the beginning of each round, one player is designated as the **small blind** and the other as the **big blind**. These roles **alternate between successive rounds**. At the start of the round, the Small Blind acts first and posts **10** chips into the pot. The Big Blind then posts **20** chips into the pot. These **forced bets** initiate the pot and leave small blind to act next.

6 Betting

After the hole cards are dealt, the **first round of betting** begins, the **small blind is the first player to act**. In this first action, they can fold, call (match the Big Blind), or raise.

In all **other rounds of betting**, the **big blind is the first person to act**. In this action, the player may check (bet 0) or bet. The minimum legal bet is one big blind, that is 20 chips. The maximum legal bet is bounded by both players' remaining stack sizes.

At any point during a betting round, a player may take one of the following actions, subject to the game state: **fold**, **check** (bet 0 chips, only if there is no wager to call), **call**, or **raise**. When a player decides to raise, the minimum raise **to** (note: raise *to*, not raise *by*) is:

$$\text{Minimum Raise} = \text{Current Wager} + \text{Cost to Call} + \max(\text{Big Blind}, \text{Cost to Call})$$

$$\text{Maximum Raise} = \min(\text{Your Chips}, \text{Opponent's Chips})$$

There is **no limit on the number of consecutive raises that may occur** in a round of betting. **The round of betting ends when a player calls, when a player folds, or when both players check in a row.**

In the first round of betting, if the small blind calls right away, the round does not end. The opponent is given the opportunity to act, they can check or raise.

If the bot makes an **invalid bet** (including negative bets, fractional bets, or betting more chips than it has), it will automatically be **assumed to have folded**.

Note: At any point during the round, a bet or raise that exceeds your opponent's remaining chips is invalid.

7 Exchange

Before every round of betting, both players participate in an **exchange of cards** (i.e, players exchange a certain number of cards with the opponent). The number of cards exchanged is 3 in the first round of exchange, 2 in the second, and 1 in the third and last one. The specific cards from a player's hole cards to be given to their opponent are chosen by the player themselves. A player is not allowed to skip an exchange round - they must participate in it and exchange exactly as many cards as mandated by the rules of the game at that stage.

8 Showdown

If both players remain in the hand after the final betting round, the hand proceeds to showdown. At showdown, **each player's best five-card hand** (Hand rankings follow the **standard Texas Hold'em hierarchy**) is evaluated using the seven hole cards they hold at that instant. The **higher-ranked hand wins the pot**, and bankrolls are updated accordingly. **In case of a tie, the pot is split equally**. If a player folds at any point before showdown, the remaining player wins the pot without revealing their cards.

9 Constraints

- Every time the engine queries the bot, it gets **2 seconds** to respond.
- The total response time (sum of response times of all queries for 1000 rounds) is limited to **20 seconds**.
- If the bot does not respond within the specified time limit, it is automatically assumed to have **folded**.